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Gate-based counterdiabatic driving with complexity guarantees

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- Counterdiabatic driving (CD)

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- Hardness of CD and need for fair comparison with adiabatic state preparation (ASP)

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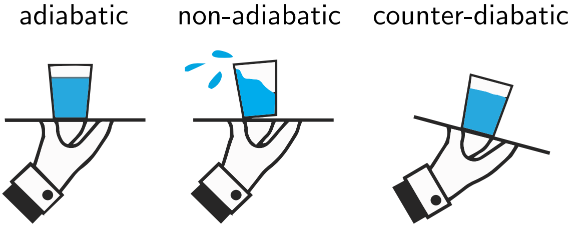

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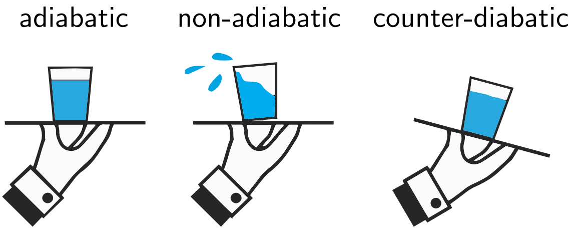

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A is known as adiabatic gauge potential (AGP)

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Works well for simple systems...

...but for complex systems, compute time scales like $\sim \dim(\mathcal{H})^2$

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- does the computational complexity needed to boost fidelity through CD scale favourably...
- ...as compared to simply increasing the evolution time in ASP?

By the way, how do we compare complexities?

We count quantum gates on fault-tolerant QC

For ASP, we know the the JRS bound

$$T \geq O(\Delta^{-3}\epsilon^{-1}) \Rightarrow |\langle n(\lambda_{\mathsf{f}})|\psi(T)\rangle|^2 \geq 1-\epsilon^2$$

where Δ is the minimum gap around $|n(\lambda)\rangle$ along the path

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To turn this into quantum gates, we trotterise

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With a k^{th}-order Lie-Suzuki-Trotter formula, we need

$$O\left(\Delta^{-(3+\frac{3}{2k})}\epsilon^{-(1+\frac{1}{k})}\,(25/3)^k\right)$$

quantum gates to obtain fidelity $\geq 1-\epsilon^2$

Gate-based counterdiabatic driving

Goal: to approximate the unitary

$$\mathrm{U}(\lambda_{\mathrm{f}}, \lambda_{\mathrm{i}}) = \sum_n |n(\lambda_{\mathrm{f}})\rangle\langle n(\lambda_{\mathrm{i}})| = \mathcal{T} \exp\left[-\mathrm{i} \int_{\lambda_{\mathrm{i}}}^{\lambda_{\mathrm{f}}} \mathrm{d}\lambda\, \mathrm{A}(\lambda))\right]$$

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We upper bound euclidean distance $\|(\mathrm{U} - \tilde{\mathrm{U}})|n(\lambda_\mathrm{i})\rangle\|$, since

$$\|(\mathrm{U} - \tilde{\mathrm{U}})|n(\lambda_\mathrm{i})\rangle\| \leq \epsilon$$

$$\Rightarrow |\langle n(\lambda_\mathrm{i})|\mathrm{U}^\dagger \tilde{\mathrm{U}}|n(\lambda_\mathrm{i})\rangle|^2 \geq 1 - \epsilon^2$$

Time-integral form of AGP

$$\mathsf{A} = \frac{1}{2}\lim_{\eta\to 0}\int_{-\infty}^{\infty} \mathrm{d}\tau\, \mathrm{e}^{-\eta|\tau|}\,\mathrm{sgn}(\tau)\,\mathrm{e}^{-\mathrm{i}\mathsf{H}\tau}\partial_\lambda \mathsf{H}\,\mathrm{e}^{\mathrm{i}\mathsf{H}\tau}$$

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has matrix elements (in energy eigenbasis)

$$\langle m|\mathsf{A}|n\rangle = \lim_{\eta\to 0}\frac{\omega_{mn}}{\mathrm{i}(\omega_{mn}^2+\eta^2)}\langle m|\partial_\lambda \mathsf{H}|n\rangle \qquad \omega_{mn} = E_m - E_n$$

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At the same time, by the Hellman-Feynman theorem,

$$\langle m|\mathrm{A}|n\rangle = \mathrm{i}\langle m|\partial_\lambda|n\rangle = \frac{\langle m|\partial_\lambda \mathrm{H}|n\rangle}{\mathrm{i}\omega_{mn}}$$

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1. Regularise and truncate integral expression:

$$\mathsf{A} \approx \mathsf{A}_{\eta,a} = \frac{1}{2}\int_{-a}^{a} \mathrm{d}\tau \, \mathrm{e}^{-\eta|\tau|} \, \mathrm{sgn}(\tau) \, \mathrm{e}^{-\mathrm{i}\mathsf{H}\tau} \partial_\lambda \mathsf{H} \, \mathrm{e}^{\mathrm{i}\mathsf{H}\tau}$$

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2. Approximate the time integral with a sum:

$$\mathsf{A}_{\eta,a}(\lambda) \approx \mathsf{A}_{\eta,a}^{M,q}(\lambda) = \sum_{\kappa=-M}^{M}\sum_{\alpha=0}^{q} \mathsf{C}_{\kappa,\alpha}(\lambda)$$

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3. Approximate time-ordered exponential

$$\mathsf{U}_{\eta,a}^{M,q}(\lambda_\mathsf{f},\lambda_\mathsf{i}) = \mathcal{T}\exp\left[-\mathrm{i}\int_{\lambda_\mathsf{i}}^{\lambda_\mathsf{f}} \mathrm{d}\lambda\, \mathsf{A}_{\eta,a}^{M,q}(\lambda)\right]$$

with a Lie-Suzuki-Trotter (LST) formula

$$\mathsf{U}_{\eta,a}^{M,q}(\lambda_\mathsf{f},\lambda_\mathsf{i}) \approx \tilde{\mathsf{U}}(\lambda_\mathsf{f},\lambda_\mathsf{i}) = \prod_j \exp[-\mathrm{i}\,\mathsf{C}_{\kappa_j,\alpha_j}(\lambda_j)\delta\lambda_j]$$

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By a triangle equality, the total error is then at most

$$\begin{aligned}\|(\mathrm{U}-\tilde{\mathrm{U}})|n(\lambda_\mathrm{i})\rangle\| \leq \|(\mathrm{U}-\mathrm{U}_{\eta,a})|n(\lambda_\mathrm{i})\rangle\| \\ + \|(\mathrm{U}_{\eta,a}-\mathrm{U}_{\eta,a}^{M,q})|n(\lambda_\mathrm{i})\rangle\| + \|(\mathrm{U}_{\eta,a}^{M,q}-\tilde{\mathrm{U}})|n(\lambda_\mathrm{i})\rangle\|\end{aligned}$$

Regularisation

$$\langle m|\mathsf{A}_{\eta,\infty}|n\rangle = \frac{\omega_{mn}}{\mathrm{i}(\omega_{mn}^2 + \eta^2)}\langle m|\partial_\lambda \mathsf{H}|n\rangle \qquad \omega_{mn} = E_m - E_n$$

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Truncation introduces an additional error $\sim \mathrm{e}^{-\eta a}$; this leads to

$$\eta \le O(\Delta^{3/2}\epsilon^{1/2}), \quad a \ge \tilde{O}(1/\eta)$$

$$\Rightarrow \|(\mathrm{U}-\mathrm{U}_{\eta,a})|n(\lambda_\mathrm{i})\rangle\| \le \epsilon$$

Riemann sum

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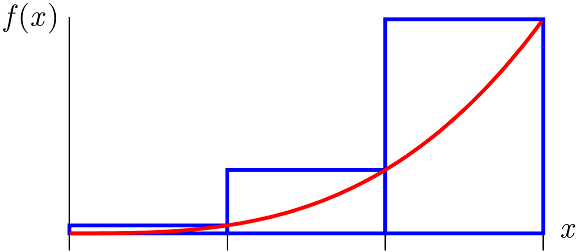

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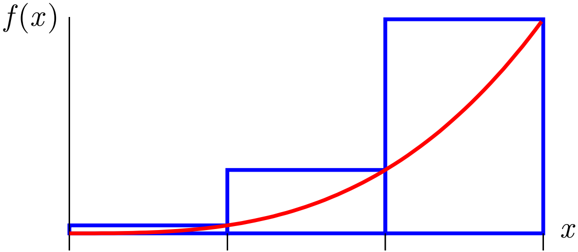

$$\mathsf{A}^{M,0}_{\eta,a} = \frac{1}{2} \sum_{\kappa=-M}^{M} \delta\tau_\kappa \, \mathrm{e}^{-\eta|\tau_\kappa|} \operatorname{sgn} \tau_\kappa \, \mathrm{e}^{-\mathrm{i}\mathsf{H}\tau_\kappa} \partial_\lambda \mathsf{H} \, \mathrm{e}^{\mathrm{i}\mathsf{H}\tau_\kappa}$$

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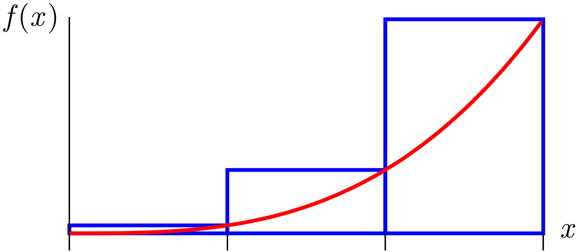

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$$M \geq O(\eta^{-2}\epsilon^{-1}) \Rightarrow \|(\mathsf{U}_{\eta,a} - \mathsf{U}^{M,0}_{\eta,a})|n(\lambda_{\mathrm{i}})\rangle\| \leq \epsilon$$

Trapezoidal rule

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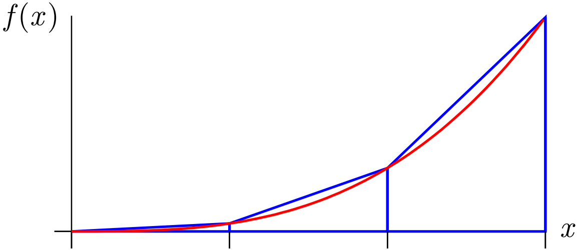

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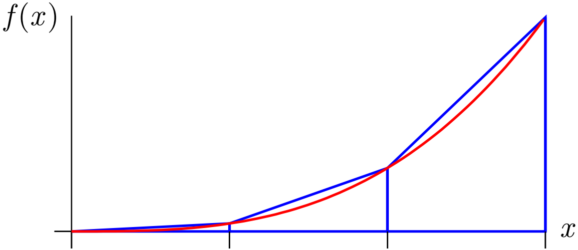

$$\mathsf{A}^{M,1}_{\eta,a} = \frac{1}{2}\sum_{\kappa=-M}^{M} \frac{\delta\tau_\kappa}{2}\left(\mathrm{e}^{-\eta|\tau_\kappa|}\,\mathrm{sgn}\,\tau_\kappa\,\mathrm{e}^{-\mathrm{i}\mathsf{H}\tau_\kappa}\partial_\lambda \mathsf{H}\,\mathrm{e}^{\mathrm{i}\mathsf{H}\tau_\kappa} + (\tau_\kappa \leftrightarrow \tau_{\kappa-1})\right)$$

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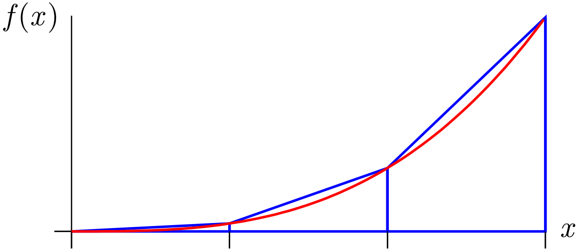

$$\mathsf{A}_{\eta,a}^{M,1} = \frac{1}{2}\sum_{\kappa=-M}^{M}\frac{\delta\tau_\kappa}{2}\left(\mathsf{e}^{-\eta|\tau_\kappa|}\,\mathsf{sgn}\,\tau_\kappa\,\mathsf{e}^{-\mathsf{i}\mathsf{H}\tau_\kappa}\partial_\lambda\mathsf{H}\,\mathsf{e}^{\mathsf{i}\mathsf{H}\tau_\kappa} + (\tau_\kappa \leftrightarrow \tau_{\kappa-1})\right)$$

$$M \geq O(\eta^{-3/2}\epsilon^{-1/2}) \Rightarrow \|(\mathsf{U}_{\eta,a} - \mathsf{U}_{\eta,a}^{M,1})|n(\lambda_\mathsf{i})\rangle\| \leq \epsilon$$

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$$\mathrm{A}_{\eta,a}^{M,q} = \frac{1}{2} \sum_{\kappa=-M}^{M} \sum_{\alpha=0}^{q} \delta\tau_\kappa w_{\kappa,\alpha} \Big(\mathrm{e}^{-\eta|\tau_{\kappa,\alpha}|} \operatorname{sgn} \tau_{\kappa,\alpha} \mathrm{e}^{-\mathrm{i}\mathrm{H}\tau_{\kappa,\alpha}} \partial_\lambda \mathrm{H} \, \mathrm{e}^{\mathrm{i}\mathrm{H}\tau_{\kappa,\alpha}}) \Big)$$

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with $\sum_{\alpha=0}^{q} w_{\kappa,\alpha} = 1$

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Weights $w_{\kappa,\alpha}$ only depend on our choice of $\tau_{\kappa,\alpha}$!

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Weights $w_{\kappa,\alpha}$ only depend on our choice of $\tau_{\kappa,\alpha}$!

In the end, we get

$$M \geq O\left(\eta^{-1-\frac{1}{q+1}} \epsilon^{-\frac{1}{q+1}} (q+1)^{-1}\right) \Rightarrow \|(\mathrm{U}_{\eta,a} - \mathrm{U}_{\eta,a}^{M,q})|n(\lambda_\mathrm{i})\rangle\| \leq \epsilon$$

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In the end, we get

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Trotter expansions serve to approximate

$$\mathcal{T}\,\mathrm{exp}[-\mathrm{i}\int_{\lambda_\mathrm{i}}^{\lambda_\mathrm{f}} \mathrm{d}\lambda \sum_{\kappa,\alpha} \mathsf{C}_{\kappa,\alpha}(\lambda)] \approx \prod_j \mathrm{exp}[-\mathrm{i}\mathsf{C}_{\kappa_j,\alpha_j}(\lambda_j)\delta\lambda_j]$$

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One can construct a k^{th}-order LST formula $\tilde{\mathsf{U}}_k$ with

$$\tilde{O}\left(\frac{M q}{\eta^{1+\frac{1}{2k}}\epsilon^{\frac{1}{2k}}}(25/3)^k\right)$$

quantum gates to guarantee

$$\|\mathcal{T}\exp[-\mathrm{i}\int_{\lambda_\mathrm{i}}^{\lambda_\mathrm{f}} \mathrm{d}\lambda \sum_{\kappa,\alpha} \mathsf{C}_{\kappa,\alpha}(\lambda)] - \tilde{\mathsf{U}}_k(\lambda_\mathrm{f},\lambda_\mathrm{i})\| \leq \epsilon$$

Putting everything together:

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$$\Rightarrow \qquad \tilde{O}\left(\frac{M q}{\eta^{1+\frac{1}{2k}} \epsilon^{\frac{1}{2k}}} (25/3)^k\right)$$
$$= \tilde{O}\left(\Delta^{-(3+\frac{3}{4k}+\frac{3}{2(q+1)})} \epsilon^{-(1+\frac{3}{4k}+\frac{3}{2(q+1)})} (25/3)^k\right)$$

to guarantee $|\langle n(\lambda_{\mathrm{f}})|\tilde{\mathrm{U}}|n(\lambda_{\mathrm{i}})\rangle|^2 \geq 1 - \epsilon^2$

So what do we have?

Gate-based ASP:

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Gate-based CD:

$$\tilde{O}\left(\Delta^{-(3+\frac{3}{4k}+\frac{3}{2(q+1)})}\epsilon^{-(1+\frac{3}{4k}+\frac{3}{2(q+1)})}\,(25/3)^k\right)$$

where q can be made big cheaply

Gate-based ASP:

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"Shortcut to adiabaticity" turns out not to be much of a shortcut after all...

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Outlook

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- Nonetheless, if quantum resources are scarce, CD with precomputed AGP can be useful
- Eigenstate-specific AGP approximations seem promising (perhaps even crucial?)

Thank you!

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